

- 25 B** Induced e.m.f. $E = B_v A f$,
where B_v is the vertical component of the Earth's field

$$\therefore E = B \sin 72^\circ \times \pi r^2 \times f$$

$$= 5.6 \times 10^{-5} \sin 72^\circ \times \pi (0.55)^2 \times 1.6 = 8.099 \times 10^{-5} \text{ V} = 81 \mu\text{V}$$

- 26 A 1st circuit:** Let supply be V